CURRICULUM VITAE

# Dr. Hoang Nguyen


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## Profile

I am a security and privacy researcher with a doctorate (Dr. sc.) from the Institute of Information Security at ETH Zurich. I develop formal methods for building software systems that are secure and privacy-compliant, with a focus on access control, privacy by design, and data protection. My work is published at leading security and privacy venues, including papers at ACM CCS and PETS.

## Education

**Doctor of Science (Dr. sc.), Computer Science** 2022 – 2026
*ETH Zurich, Institute of Information Security* Switzerland

**Master of Science, Formal Methods in Computer Science** 2020 – 2021
*Universidad Autónoma | Complutense | Politécnica de Madrid* Spain

**Bachelor of Science, Computer Science** 2014 – 2018
*Vietnamese-German University | Frankfurt University of Applied Sciences* Vietnam | Germany

## Employment

**Research and Teaching Assistant** Sep 2026 – Present
*Vietnamese-German University, Faculty of Engineering* Vietnam

**Doctoral Assistant** Jul 2022 – Aug 2026
*ETH Zurich, Institute of Information Security* Switzerland

**Research Intern** Feb 2022 – Jun 2022
*IMDEA Software Institute* Spain

**Research Assistant** Aug 2021 – Jan 2022
*DAAD German Academic Exchange Service* Vietnam

**Technical Research Assistant** Feb 2019 – Aug 2020
*Vietnamese-German University, Faculty of Engineering* Vietnam

**Software Engineer** Feb 2018 – Jan 2019
*Axon Active* Vietnam

## Awards and Scholarships

**Vingroup Graduate Scholarship** 2022 – 2026
*Vingroup Science and Technology Scholarship Program* Vietnam

**DAAD Exchange Scholarship** 2016
*DAAD German Academic Exchange Service* Germany

**Merit Scholarship for Excellent Studies** 2014 – 2018
*Vietnamese-German University* Vietnam

## Skills

- **Security and privacy:** access control, data protection, privacy compliance
- **Formal methods and modeling:** model-driven engineering, UML/OCL, domain-specific languages, model transformations, code generation, logics and formal languages
- **Languages:** Vietnamese (native), English (professional), German (basic)

## Research Output

### Research Interest

My research lies at the intersection of security, privacy, and formal methods. I develop model-driven methodologies for building software systems that enforce security and privacy requirements by construction, with an emphasis on data protection: controlling who can access data, and ensuring that data is used only for the purposes for which it was collected.

**Fine-Grained Access Control** In the context of model-driven engineering, [12] discusses some of the challenges and opportunities of the mapping from the Object Constraint Language (OCL) to the Structured Query Language (SQL). [11] defines a novel mapping, called OCL2PSQL, that overcomes some of the aforementioned challenges. [8] proposes the definition of this OCL2PSQL mapping as a case to study the readiness of the state-of-the-art transformation tools. [10] formally characterizes the enforcement of Fine-Grained Access Control (FGAC) policies for a subset of SQL queries. [9] then proposes a model-based methodology for developing an FGAC enforcement mechanism following our formal characterization. [6, 7, 14, 15] extend the methodology on several dimensions, namely, performance, expressiveness, and correctness.

**Privacy and Data Protection** We propose a model-driven development methodology that incorporates privacy policies into the system design [1]. In a nutshell, software engineers design a privacy model that supports notions of *purpose limitation*, *data subject consent* [4] and rely on predefined code generation to generate correct-by-design privacy enforcement [3]. We conduct a large-scale empirical study asking developers to implement privacy requirements to showcase the effectiveness of our approach compared to the code-centric development techniques [2, 5].

### Publications

1. **Hoàng Nguyễn**.
   Model-driven Privacy.
   *Doctoral Thesis*, ETH Zurich, 2026.

2. **Hoàng Nguyễn**, Srđan Krstić, and David Basin.
   Models Trump Code: An Empirical Analysis of GDPR Compliance.
   *ACM Conference on Computer and Communications Security* (CCS), 2026.

3. David Basin, François Hublet, Srđan Krstić, and **Hoàng Nguyễn**.
   Mechanizing Privacy by Design.
   *ACM Conference on Computer and Communications Security* (CCS), 2025.

4. Srđan Krstić, **Hoàng Nguyễn**, and David Basin.
   Model-driven Privacy.
   *Privacy Enhancing Technologies Symposium* (PETS), 2024.

5. David Basin, Juan Guarnizo, Srđan Krstić, **Hoàng Nguyễn**, and Martin Ochoa.
   Is Modeling Access Control Worth It?
   *ACM Conference on Computer and Communications Security* (CCS), 2023.

6. **Hoàng Nguyễn** and Manuel Clavel.
   An Extended Model-based Characterization of Fine-Grained Access Control for SQL Queries.
   Foundations Track, *Model Driven Engineering Languages and Systems* (MoDELS), 2023.

7. **Hoàng Nguyễn**.
   Intelligent Enforcement of Fine-Grained Access Control Policies for SQL Queries.
   *Master's Thesis*, Universidad Autónoma de Madrid, 2021.

8. **Hoàng Nguyễn**, Antonio García-Domínguez, and Manuel Clavel.
   The Transformation Tools Contest 2021: OCL2PSQL case.
   *Software Technologies: Applications and Foundations* (STAF), 3089:1–11, 2021.

9. **Hoàng Nguyễn** and Manuel Clavel.
   A Model-Driven Approach for Enforcing Fine-Grained Access Control for SQL Queries.
   *SN Computer Science* (SNCS), 2(5):370–389, 2021.

10. **Hoàng Nguyễn** and Manuel Clavel.
    Model-based Characterization of Fine-Grained Access Control Authorization for SQL Queries.
    *Journal of Object Technology* (JOT), 19(3):1–13, 2020.

11. **Hoàng Nguyễn** and Manuel Clavel.
    OCL2PSQL: An OCL-to-SQL Code-Generator for Model-Driven Engineering.
    *Future Data and Security Engineering* (FDSE), 11814:185–203, 2019.

12. Manuel Clavel and **Hoàng Nguyễn**.
    Mapping OCL into SQL: Challenges and Opportunities Ahead.
    OCL Workshop, *Model Driven Engineering Languages and Systems* (MoDELS), 2513:3–16, 2019.

13. **Hoàng Nguyễn**.
    On the Critical Configurations of the Chip Firing Game.
    *Bachelor's Thesis*, Vietnamese-German University, 2018.

## Drafts

14. **Hoàng Nguyễn** and Manuel Clavel.
    Proving Correctness for SQL Implementations of OCL Constraints.
    Draft version, arXiv, 2024.

15. **Hoàng Nguyễn** and Manuel Clavel.
    Optimising Fine-Grained Access Control Policy Enforcement for Database Queries.
    Draft version, arXiv, 2022.

## Artifacts and Tools

16. **Hoàng Nguyễn**, Srđan Krstić, and David Basin.
    MDP: Model-driven Privacy tools.
    https://nuactiongui.github.io

17. **Hoàng Nguyễn** and Manuel Clavel.
    OCL2PSQL Mapping.
    https://github.com/MoDELSVGU/OCL2PSQL

18. **Hoàng Nguyễn** and Manuel Clavel.

SQLSI: SQL Security Injector.
https://github.com/MoDELSVGU/SQLSI

19. **Hoàng Nguyễn** and Manuel Clavel.
SQL2MSFOL: SQL statements to Many-Sorted First-Order Logic.
https://github.com/MoDELSVGU/SQL2MSFOL

## Teaching Activities

**Real-Time Systems** — Spring 2027 (upcoming)
*Bachelor's Course in Computer Science, Vietnamese-German University* — Vietnam
– Lecturer (Spring 2027)

**IT Security** — Fall 2026
*Bachelor's Course in Computer Science, Vietnamese-German University* — Vietnam
– Lab Assistant (Fall 2026)

**Information Security** — Spring 2023 – Spring 2025
*Bachelor's Course in Computer Science, ETH Zurich* — Switzerland
– Teaching Assistant (Spring 2023, Spring 2024, Spring 2025)

**Security Engineering** — Fall 2022 – Fall 2025
*Master's Course in Computer Science, ETH Zurich* — Switzerland
– Head of the Teaching Assistants (Fall 2023, Fall 2024)
– Teaching Assistant (Fall 2022, Fall 2025)

## Supervision Activities

**Misconception-Aware Interactive Python Learning for Beginners** — Ongoing
*Ms. Lê Minh Bảo Ngọc, Bachelor's Thesis, Vietnamese-German University* — Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

**Formal Reasoning about Data Protection** — Completed
*Mr. Andrei Cotor, Master's Thesis, ETH Zurich* — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

**GDPR-Compliant Delegated Visitor Access Control for Campus Gate-Entry** — Completed
*Mr. Dương Võ Thiên Bảo, Bachelor's Thesis, Vietnamese-German University* — Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel and Dr. Lê Lam Sơn

**$\nu$ActionGUI: Process Permissions and More Data Subject Rights** — Completed
*Mr. Andrei Cotor, Master's Semester Project, ETH Zurich* — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

**Privacy-Aware Object-Relational Mappers** — Completed
*Ms. Sofie Daniels, Master's Thesis, ETH Zurich* — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

**$\nu$ActionGUI** — Completed
*Mr. Raymond Nasr, Master's Semester Project, ETH Zurich* — Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

**Automated Threat Modeling for Industrial Control Systems** — Completed
*Mr. Mehdi Berrada, Master's Semester Project, ETH Zurich* — Switzerland

**Model Transformations for Privacy-by-Design** Completed
*Mr. Daniel Gradwohl, Bachelor's Thesis, ETH Zurich* Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

**Secure Object-Relational Mappers** Completed
*Mr. Karlo Piskor, Bachelor's Thesis, ETH Zurich* Switzerland
– Co-supervised with Assoc. Prof. Dr. Srđan Krstić

**Fine-Grained Access Control in NoSQL** Completed
*Ms. Lê Minh Thư, Bachelor's Thesis, Vietnamese-German University* Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

**A Web-Based Validator for Data Constraints** Completed
*Ms. Phạm Hồ Linh Thư, Bachelor's Thesis, Vietnamese-German University* Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

**A Web-Based Intelligent Editor for Security Modeling** Completed
*Mr. Hoàng Thái Dương, Bachelor's Thesis, Vietnamese-German University* Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

**Intelligent Web-Based Editor for Data Modeling** Completed
*Mr. Lữ Đình Tuấn, Bachelor's Thesis, Vietnamese-German University* Vietnam
– Co-supervised with Assoc. Prof. Dr. Manuel Clavel

## Voluntary Activities

**Master Trainer** Sep 2026 – Present
*ASEAN Foundation, SCAM Ready ASEAN* Vietnam
– Training communities to identify and prevent online scams

**Mentor** Apr 2024 – Dec 2024
*Vietnamese-German University, Student-Alumni Mentoring Program* Vietnam
– Mentoring computer science undergraduates

**Translator** Feb 2019 – Apr 2019
*Như Chưa Hề Có Cuộc Chia Ly* Vietnam
– Volunteer translation of program materials to reach international audiences